



IdeaVault

Complete Setup Guide — Windows

From zero to running application
MongoDB + Node.js + React
Version 1.0 — July 2026

Table of Contents

1. Prerequisites — What You Need
2. Install Node.js
3. Install Git
4. Install MongoDB + MongoDB Compass
5. Clone the Repository
6. Install Dependencies
7. Configure Environment Variables
8. Start MongoDB
9. Seed the Database
10. Run the Application
11. Port Reference
12. YouTube Tutorials
13. Troubleshooting

1. Prerequisites — What You Need

Software	Version	Purpose	Download
Node.js	18+ (LTS recommended)	JavaScript runtime for the server	nodejs.org
npm	9+ (comes with Node.js)	Package manager	Included with Node.js
Git	Latest	Version control, clone the repo	git-scm.com
MongoDB	7.0+	Database server	mongodb.com
MongoDB Compass	Latest	GUI for MongoDB (optional but recommended)	mongodb.com
Code Editor	—	VS Code recommended	code.visualstudio.com

Tip: You can check if Node.js and npm are installed by opening a terminal and running `node --version` and `npm --version`.

2. Install Node.js

Step 1: Download Node.js

Go to <https://nodejs.org> and download the **LTS (Long Term Support)** version for Windows.

The installer will be named something like `node-v20.x.x-x64.msi`.

Step 2: Run the Installer

- Double-click the downloaded .msi file
- Click "Next" through the wizard
- **Important:** On the "Tools for Native Modules" screen, you may optionally check the box to install necessary tools — this helps with some npm packages
- Click "Install" and finish the wizard

Step 3: Verify Installation

Open a new terminal (Command Prompt or PowerShell) and run:

```
node --version
# Should output something like: v20.15.1

npm --version
# Should output something like: 10.7.0
```

Note: If you see "'node' is not recognized", close and reopen your terminal. If it still doesn't work, you may need to add Node.js to your PATH manually.

3. Install Git

Step 1: Download Git

Go to <https://git-scm.com> and download the Windows installer.

Step 2: Install Git

- Run the installer
- Accept the defaults (they work well for most people)
- Make sure "Git from the command line and also from 3rd-party software" is selected

Step 3: Verify

```
git --version  
# Should output something like: git version 2.45.0.windows.1
```

4. Install MongoDB + MongoDB Compass

4.1 Install MongoDB Server

Step 1: Download MongoDB Community Edition

Go to <https://www.mongodb.com/try/download/community>.

- Select **Version**: 7.0.x (or latest)
- Select **Platform**: Windows
- Select **Package**: MSI
- Click "Download"

Step 2: Run the Installer

- Double-click the downloaded .msi file
- Click "Next" through the wizard
- On the "Setup Type" screen, choose **"Complete"**
- **Important**: On the "Service Configuration" screen, keep "Install MongoDB as a Service" checked — this starts MongoDB automatically
- On the "MongoDB Compass" screen, you can uncheck it if you want to install Compass separately (see below)
- Click "Install" and finish the wizard

Step 3: Verify MongoDB is Running

Open a new terminal and run:

```
mongod --version  
# Should output something like: db version v7.0.x
```

MongoDB should be running as a Windows service. You can check in Services (Win+R → services.msc) that "MongoDB Server" is running.

4.2 Install MongoDB Compass

Step 1: Download Compass

Go to <https://www.mongodb.com/try/download/compass>

Download the Windows .exe installer.

Step 2: Install and Connect

- Run the downloaded .exe file
- Click "Install"
- When Compass opens, the connection string should be pre-filled: `mongodb://localhost:27017`
- Click **"Connect"**
- You should see your MongoDB databases in the left sidebar

Tip: MongoDB Compass lets you visually browse collections, view documents, and run queries without using the command line.

4.3 Verify with MongoDB Shell (Optional)

Alternative: mongosh

If you prefer the command line, install `mongosh` (MongoDB Shell):

```
# In PowerShell
mongosh

# You should see:
# Current Mongosh Log ID: ...
# Connecting to: mongodb://127.0.0.1:27017
# Using MongoDB: 7.0.x
```

5. Clone the Repository

Step 1: Open a Terminal

Open PowerShell or Command Prompt and navigate to where you want the project:

```
# Example: navigate to your projects folder  
cd C:\Users\YourName\projects
```

Step 2: Clone

```
git clone https://github.com/AimalShah/idea-vault.git  
cd idea-vault
```

Step 3: Verify

```
dir  
# Should show:  
# .git/  
# CONTEXT.md/  
# docs/  
# client/  
# server/  
# package.json
```


6. Install Dependencies

You need to install dependencies for three packages: root, server, and client.

Step 1: Install Root Dependencies

```
# From the idea-vault/ directory
npm install
# Installs: concurrently
```

Step 2: Install Server Dependencies

```
npm install --prefix server
# Installs: express, mongoose, bcryptjs, jsonwebtoken, cookie-parser, cors, dotenv
# Dev deps: supertest
```

Step 3: Install Client Dependencies

```
npm install --prefix client
# Installs: react, react-dom, react-router-dom, axios, lucide-react
# Dev deps: vite, @vitejs/plugin-react, oxlint
```

Tip: You can install all three at once with a single command from the root:

```
npm install && npm install --prefix server && npm install --prefix client
```

7. Configure Environment Variables

7.1 Server .env File

Step 1: Check for .env file

The server needs a `.env` file in the `server/` directory. Check if it exists:

```
dir server\.env
```

Step 2: Create .env (if missing)

If the file doesn't exist, create it with the following content:

```
# server/.env
MONGO_URI=mongodb://localhost:27017/idea-vault
JWT_SECRET=idea-vault-dev-secret-key
PORT=5000
```

Step 3: Customize (Optional)

Variable	Default	Description
MONGO_URI	<code>mongodb://localhost:27017/idea-vault</code>	MongoDB connection string. Change if using a remote MongoDB or different port.
JWT_SECRET	<code>idea-vault-dev-secret-key</code>	Secret key for signing JWT tokens. Change this in production!
PORT	<code>5000</code>	Port for the Express server. Don't change unless 5000 is in use.

Warning: Never commit your `.env` file to git. It's already in `.gitignore`.

8. Start MongoDB

Option A: MongoDB as Windows Service (Recommended)

If you installed MongoDB with "Install as Service" checked, MongoDB starts automatically. To verify:

```
# Open Services (Win+R → services.msc)
# Look for "MongoDB Server" – status should be "Running"

# Or check in PowerShell:
Get-Service MongoDB
```

Option B: Start MongoDB Manually

```
# In PowerShell
mongod --dbpath "C:\data\db"

# Or if mongod is not in PATH:
"C:\Program Files\MongoDB\Server\7.0\bin\mongod.exe" --dbpath "C:\data\db"
```

Option C: Using MongoDB Compass

Open MongoDB Compass and connect to `mongodb://localhost:27017`. This will confirm MongoDB is running.

Verify Connection

```
# In PowerShell (if mongosh is installed)
mongosh --eval "db.runCommand({ ping: 1 })"

# Should output: { ok: 1 }
```

Note: The application expects MongoDB on port 27017 (the default). If you changed this, update `MONGO_URI` in `server/.env`.

9. Seed the Database

The seed script creates sample data so you can explore the app immediately.

Step 1: Run the Seed Script

```
# From the project root
npm run seed

# Or from the server directory
npm run seed --prefix server
```

Step 2: Verify in MongoDB Compass

Open MongoDB Compass and connect. You should see:

Collection	Documents	Content
users	3	alice@example.com, bob@example.com, carol@example.com
ideas	10	Various sample ideas
votes	~30-50	Random votes from users
comments	~10-30	1-3 comments per idea

Step 3: Test Credentials

All seed users have the same password:

```
Email: alice@example.com   Password: password123
Email: bob@example.com     Password: password123
Email: carol@example.com   Password: password123
```

Note: The seed script clears all existing data before creating new seed data. Run it again anytime to reset.

10. Run the Application

Step 1: Start Both Server and Client

```
# From the project root  
npm run dev
```

This starts two processes simultaneously:

- **Server** on `http://localhost:5000` (Express API)
- **Client** on `http://localhost:3000` (React/Vite dev server)

Step 2: Open the Application

Open your browser and go to:

```
http://localhost:3000
```

You should see the IdeaVault login page.

Step 3: Register or Log In

- Click "Sign up" to create a new account
- Or use one of the seed accounts (see Step 9)

Running Separately

If you want to run server and client independently:

```
# Terminal 1 – Server  
npm run server  
  
# Terminal 2 – Client  
npm run client
```

Running Tests

```
# From the project root  
npm test  
  
# Or from the server directory  
npm test --prefix server
```

Expected output: 34 tests passing.

11. Port Reference

Port	Service	URL	Notes
3000	Vite Dev Server (Client)	http://localhost:3000	React frontend. Only accessible when <code>npm run client</code> or <code>npm run dev</code> is running.
5000	Express Server (API)	http://localhost:5000	REST API. Proxied through Vite on port 3000 for <code>/api/*</code> requests.
27017	MongoDB	mongodb://localhost:27017	Database server. Must be running before starting the Express server.

Note: In the browser, you only access port 3000. Vite proxies all `/api/*` requests to port 5000 automatically. You do NOT need to access port 5000 directly in the browser.

12. YouTube Tutorials

Helpful video tutorials for setting up each technology:

MERN Stack Basics



MERN Stack Tutorial for Beginners — Full Course

FreeCodeCamp — Complete MERN stack walkthrough (React, Express, MongoDB, Node.js)

[Watch on YouTube](#)



Build a MERN Stack App from Scratch

Traversy Media — Step-by-step MERN project with authentication

[Watch on YouTube](#)

MongoDB Setup



MongoDB Crash Course

Traversy Media — MongoDB basics, installation, and CRUD operations

[Watch on YouTube](#)



MongoDB Compass Tutorial — GUI for MongoDB

How to use MongoDB Compass to view and manage your databases visually

[Watch on YouTube](#)



Install MongoDB on Windows — Complete Guide

Step-by-step Windows installation with MongoDB Compass setup

[Watch on YouTube](#)

Node.js & Express



Node.js and Express.js Full Course

FreeCodeCamp — Complete backend development with Node.js and Express

[Watch on YouTube](#)



Express.js Crash Course

Traversy Media — Routes, middleware, and REST APIs

[Watch on YouTube](#)

React



React Tutorial for Beginners

Programming with Mosh — Complete React fundamentals

[Watch on YouTube](#)

React Router v7 — Complete Tutorial

Learn client-side routing with React Router

[Watch on YouTube](#)

Authentication & JWT



JWT Authentication Tutorial — Node.js

How JSON Web Tokens work with httpOnly cookies

[Watch on YouTube](#)

Complete Guide to Authentication with React

Context API, protected routes, and session management

[Watch on YouTube](#)

Git & GitHub



Git & GitHub Crash Course for Beginners

Traversy Media — Essential Git commands and GitHub workflow

[Watch on YouTube](#)

13. Troubleshooting

Problem	Cause	Solution
"MongoDB connection failed" or "MongooseServerSelectionError"	MongoDB is not running	Open Services (Win+R → services.msc), find "MongoDB Server", right-click → Start. Or run <code>mongod</code> in a terminal.
"EADDRINUSE: address already in use :::5000"	Another process is using port 5000	Find the process: <code>netstat -ano findstr :5000</code> . Kill it or change PORT in <code>server/.env</code> .
"EADDRINUSE: address already in use :::3000"	Another process is using port 3000	Kill the process or close other Vite dev servers.
Page loads but "Cannot GET /api/..." errors	Vite proxy not working or server not running	Ensure both <code>npm run server</code> and <code>npm run client</code> are running. Check <code>vite.config.js</code> has the proxy configured.
"Module not found" or "Cannot find module"	Dependencies not installed	Run <code>npm install</code> in the relevant directory (root, server, or client).
"SyntaxError: Cannot use import statement outside a module"	Package.json missing <code>"type": "module"</code>	Ensure <code>server/package.json</code> has <code>"type": "module"</code> .
"nodemon is not recognized" or dev server won't start	Using <code>npm run dev</code> in server directory directly	Use <code>node --watch index.js</code> instead (the project uses Node's native watch mode, not nodemon).
"Vite: Bus error (core dumped)" on startup	Node.js version incompatibility with Vite 8	This project uses Vite 5, which should work. If you see this, check <code>node --version</code> and ensure it's v18+.
"Password incorrect" when logging in with seed data	Database wasn't seeded or wrong email	Run <code>npm run seed</code> again. Use exact emails: <code>alice@</code> , <code>bob@</code> , <code>carol@example.com</code> with password: <code>password123</code>
"Email already in use" when registering	User already exists in database	Use a different email, or run <code>npm run seed</code> to reset the database, or delete the user via MongoDB Compass.
Tests failing with "Cannot read properties of null"	Test database has leftover data from previous runs	Tests use separate databases (auth, ideas, votes, comments). If persistent, drop the test databases in MongoDB Compass.
Browser shows blank page or "localhost refused to connect"	Client dev server not running	Run <code>npm run client</code> or <code>npm run dev</code> from root. Wait for "Local: http://localhost:3000" message.

Quick Diagnostic Commands

```
# Check Node.js version
node --version

# Check if MongoDB is running (Windows)
Get-Service MongoDB

# Check if port 5000 is in use
netstat -ano | findstr :5000
```

```
# Check if port 3000 is in use
netstat -ano | findstr :3000

# Test server health
Invoke-WebRequest -Uri http://localhost:5000/api/health

# Test client
Invoke-WebRequest -Uri http://localhost:3000

# Drop test databases (if tests are stuck)
mongosh --eval "db.getSiblingDB('idea-vault-test-auth').dropDatabase()"
```

Still stuck? Check the terminal output for error messages. The server logs connection errors, and the client logs compilation errors. Most issues are caused by MongoDB not running or missing dependencies.